

DUET: A Dual-System Multi-Agent Framework for Engineering Education

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Abstract—Existing LLM-based educational systems often separate conceptual tutoring from practical experimentation and provide limited support for verifiable numerical reasoning, learner adaptation, and guidance grounded in the learner’s current experimental state. To address these limitations, this paper proposes DUET, a domain-adaptable dual-system multi-agent framework for engineering education. The tutoring system integrates learner modeling, knowledge retrieval, multi-turn context handling, and deterministic numerical verification, while the experimentation system uses specialist agents and canvas context to support experiment construction, diagnosis, and result interpretation. The framework was implemented in an undergraduate Principles of Communication course and evaluated through tutoring and experimentation tasks. The results show that numerical verification and dialogue context improve tutoring reliability, multi-agent orchestration strengthens pedagogical organization, and canvas access enhances contextual grounding and diagnostic specificity. These benefits are accompanied by additional response latency.

Keywords—multi-agent systems, intelligent tutoring, engineering education, virtual laboratories, formative feedback

I. INTRODUCTION

Large language models (LLMs) have demonstrated considerable potential for educational applications by providing natural-language explanations, worked examples, and immediate feedback. However, many LLM-based tutoring systems still treat learning support primarily as an isolated text-generation task. They may lack persistent learner modeling, independently verifiable numerical reasoning, structured domain grounding, and direct awareness of hands-on learning environments. These limitations are particularly important in engineering and STEM education, where learners must connect abstract knowledge with executable procedures and observable phenomena.

A learner may understand a mathematical model or scientific principle while remaining unable to construct a valid experi-

ment, configure its parameters, diagnose structural problems, or interpret its results. Conversely, an executable experiment may be completed without a clear understanding of the relationships among theoretical assumptions, parameter settings, and observed outcomes. Existing conversational tutors can provide general explanations, but without access to the current experimental state, their guidance cannot reliably refer to the actual components, connections, parameters, diagnostics, or results involved in the learner’s activity. This separation between knowledge-oriented tutoring and practical experimentation limits the ability of existing systems to support an integrated cycle of explanation, construction, observation, diagnosis, and reflection.

To address these challenges, this paper proposes DUET, a domain-adaptable dual-system multi-agent framework for personalized tutoring and canvas-grounded experimental inquiry. The two systems are pedagogically complementary but operationally independent. The intelligent tutoring system supports conceptual explanation and problem solving through specialized agents for learner modeling, knowledge retrieval, structured reasoning, numerical verification, and resource recommendation. The interactive experimentation system uses the current canvas context to support experiment construction, structural diagnosis, parameter analysis, simulation execution, and result interpretation. The framework separates domain-independent mechanisms, such as orchestration, learner modeling, verification, and canvas-state representation, from replaceable domain resources, including knowledge bases, specialist-agent instructions, experiment modules, and execution functions.

The main contributions of this work are threefold. First, we introduce a domain-adaptable dual-system educational architecture that supports knowledge-oriented tutoring and practical inquiry without requiring shared operational state between the two systems. Second, we extend language-model generation with system-level educational capabilities, including persistent learner modeling, knowledge grounding, deterministic numerical verification, multi-turn context handling, and canvas-aware

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diagnosis. Third, we present a capability-oriented evaluation that considers both functional availability and operational effectiveness, covering tutoring support, canvas-grounded guidance, diagnostic quality, pedagogical support, and response latency. The framework is instantiated and evaluated in an undergraduate Principles of Communication course as a representative engineering-education case; the present evaluation therefore examines the effectiveness of this domain-specific implementation rather than cross-domain generalization.

The remainder of this paper reviews related work, presents the proposed framework and its two constituent systems, reports the evaluation results, and discusses the limitations and future directions of the study.

II. RELATED WORK

LLMs have enabled flexible dialogue, personalized explanation, and instructional content generation in intelligent tutoring systems. However, their educational use also raises concerns regarding factual reliability, pedagogical control, and inappropriate answer disclosure [1]. ChatTutor organizes LLM components for course planning, instruction, reflection, and assessment [2], Chowdhury et al. combines language generation with a predefined pedagogical state structure [3]. More recent multi-agent frameworks further decompose educational tasks among specialized agents. MAIC assigns different instructional roles to multiple agents for online learning [4], whereas GenMentor coordinates agents for goal analysis, skill-gap identification, learning-path planning, and personalized content generation [5]. These studies demonstrate the benefits of structured and specialized workflows, but engineering-oriented tutoring additionally requires reliable numerical computation, answer verification, learner adaptation, and coordination with multimodal resources.

Virtual laboratories provide accessible and repeatable environments in which learners can construct experiments, configure parameters, and observe system behavior [6, 8]. Existing virtual-laboratory analytics commonly rely on interaction logs and have not fully realized their potential for real-time instructional support [7]. In particular, conventional systems generally provide predefined procedures or technical error messages rather than intelligent guidance grounded in the current experiment structure, parameter configuration, diagnostics, and simulation results. Few studies jointly address deterministic verification for engineering problems and context-aware experimental guidance. Therefore, this work develops two operationally independent but pedagogically complementary systems: a multi-agent intelligent tutoring system for knowledge construction and problem solving, and a canvas-aware multi-agent experimentation system for practical inquiry.

III. PROPOSED FRAMEWORK AND METHODS

A. Dual-System Educational Framework

As illustrated in Fig. 1, the proposed framework consists of

two operationally independent but pedagogically complementary multi-agent services. The intelligent tutoring system supports conceptual explanation, problem solving, learner modeling, knowledge retrieval, numerical verification, and learning-resource recommendation. The interactive experimentation system supports experiment construction, structural diagnosis, simulation execution, and result interpretation using the learner's current canvas state.

Each system maintains its own orchestrator, specialist agents, context representation, and execution workflow, and therefore does not require shared runtime state with the other system. The orchestration, learner-modeling, verification, and canvas-representation mechanisms are designed to be domain adaptable, while the knowledge base, specialist-agent instructions, experiment modules, and execution functions can be replaced for different engineering and STEM subjects. Learners may use either system independently or move between them during a learning activity to connect conceptual understanding with practical inquiry.

B. Multi-Agent Intelligent Tutoring System

The intelligent tutoring system adopts hierarchical routing and specialized-agent collaboration. At interaction turn t , a routing module analyzes the learner query q_t together with the dialogue history H_t and produces

$$(r_t, a_t, \Gamma_t) = \mathcal{R}(q_t, H_t), \quad (1)$$

where r_t indicates whether the query is a context-dependent follow-up, a_t denotes the selected specialist agent, and Γ_t specifies instructional settings such as response depth, language, and pedagogical style. Based on this decision, requests are assigned to agents or supporting services for conceptual explanation, problem solving, knowledge retrieval, coding support, and multimodal resource generation.

For knowledge-oriented tutoring, a solver agent coordinates the following multi-stage workflow:

$$\begin{aligned} Z_t &= \mathcal{E}(q_t, H_t), \\ K_t &= \mathcal{G}(Z_t), \\ \hat{Y}_t &= \mathcal{S}(q_t, Z_t, K_t, P_t, H_t), \\ \Omega_t &= \mathcal{V}(q_t, \hat{Y}_t, Z_t), \\ Y_t &= \mathcal{F}(\hat{Y}_t, \Omega_t). \end{aligned} \quad (2)$$

Here, $\mathcal{E}$ extracts the question type, learning objective, relevant concepts, known conditions, numerical parameters, and follow-up information into a structured task representation Z_t . The knowledge-enhancement function $\mathcal{G}$ retrieves definitions, prerequisite relations, and related concepts from the domain knowledge graph and supporting retrieval services. The learner profile P_t stores concept-familiarity estimates and interaction preferences, which are used by $\mathcal{S}$ to regulate explanation depth, terminology, examples, derivations, and response organization.

Dual-System Learning Support Framework for Engineering and STEM Education

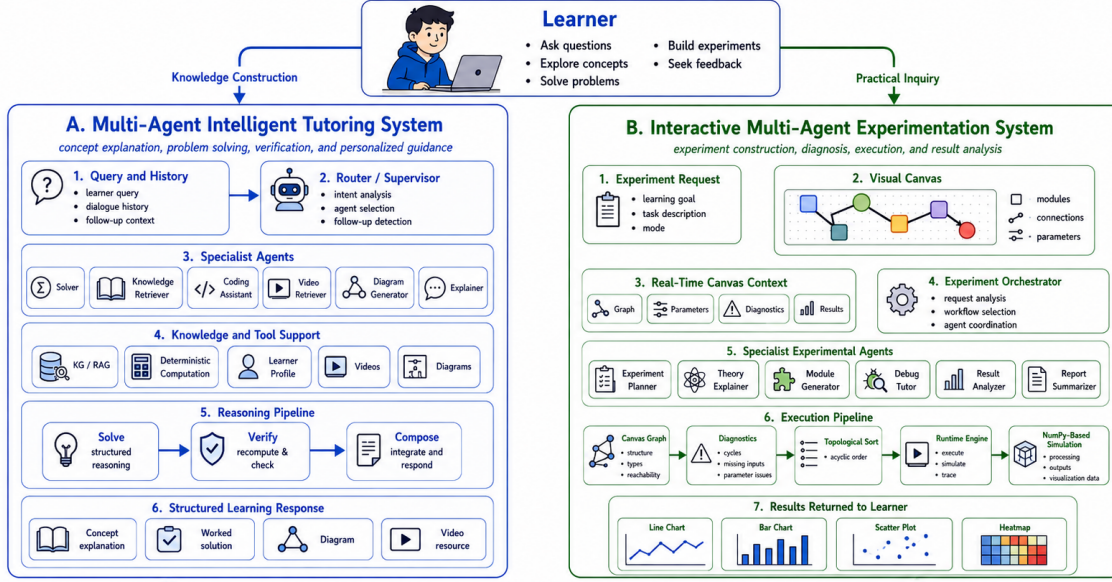


Figure 1: Overall architecture of the proposed domain-adaptable dual-system multi-agent learning framework. The intelligent tutoring system supports knowledge-oriented learning, while the interactive experimentation system provides canvas-grounded support for practical inquiry.

The solver first generates an instructional draft $\hat{Y}_t$ containing the relevant concepts, reasoning steps, formulas, and calculations. The verification component $\mathcal{V}$ is designed to extract computable expressions and independently recompute their values in a restricted deterministic Python environment. Numerical discrepancies exceeding a 1% relative-error threshold are returned to the response composer. Concept coverage, derivation completeness, unit consistency, and instructional organization are reviewed in a separate critique stage. The finalization function $\mathcal{F}$ uses these signals to revise and organize the final response Y_t .

Learner information is updated from explicit evidence identified in learner–system interactions. For a learner and concept, the new evidence weight, temporally decayed previous evidence, and bounded familiarity update are expressed as

$$\begin{aligned} w_t &= \omega_t e_t c_t, \\ \tilde{n}_{t-1} &= n_{t-1} \exp\left(-\frac{\Delta t}{\tau}\right), \\ m_t &= \text{clip}\left(\frac{m_{t-1}\tilde{n}_{t-1} + s_t w_t}{\tilde{n}_{t-1} + w_t}, m_{t-1} - \delta, m_{t-1} + \delta\right). \end{aligned} \quad (3)$$

Here, ω_t , e_t , and c_t denote source reliability, evidence strength, and observation confidence, respectively; s_t is the newly observed score; τ controls temporal decay; and δ limits the effect of a single interaction. The resulting profile is converted into instructional context and used to adapt the response to the learner’s current familiarity and preferences.

When appropriate, the orchestration workflow supplements

the response with retrieved educational videos, existing instructional diagrams, or generated visual explanations. The final output therefore combines structured reasoning, knowledge support, verification signals, learner information, dialogue context, and selected multimodal resources.

C. Interactive Multi-Agent Experimentation System

The interactive experimentation system provides a visual and executable environment in which learners construct workflows by placing, connecting, and configuring functional modules. They can modify parameters, run simulations, and inspect numerical or visual results. At time t , the experiment and its current instructional context are represented as

$$\begin{aligned} G_t &= (V_t, \mathcal{A}_t), \\ C_t &= \{G_t, \Theta_t, D_t, R_t\}, \end{aligned} \quad (4)$$

where V_t is the set of modules, $\mathcal{A}_t$ is the set of directed signal-flow connections, Θ_t contains module parameters, D_t contains structural and execution diagnostics, and R_t contains the latest simulation results. The context is updated whenever a learner adds a module, changes a connection, modifies a parameter, or executes the experiment.

A high-level experiment orchestrator jointly analyzes the learner’s request, the current canvas context, and the experiment-session history:

$$a_t^{\text{exp}} = \pi_{\text{exp}}(q_t, C_t, H_t^{\text{exp}}), \quad (5)$$

where π_{exp} denotes the orchestration policy and a_t^{exp} denotes the selected specialist agent or multi-agent workflow. The or-

chestrator may provide direct feedback, invoke a single specialist, organize a sequential workflow, or request clarification. Specialist agents support experiment planning, theoretical explanation, module construction, error diagnosis, result interpretation, and report generation. For complex tasks, learner checkpoints can be introduced so that students participate in construction, parameter adjustment, and execution rather than receiving a fully generated experiment.

At the execution layer, modules are processed in topological order. For a module v , its output is abstractly expressed as

$$y_v = f_v(\{y_u \mid (u, v) \in \mathcal{A}_t\}, \theta_v), \quad (6)$$

where f_v is the processing function of module v and θ_v contains its configured parameters. The implemented NumPy backend supports communication-engineering operations such as signal generation, modulation, filtering, channel simulation, demodulation, and result observation. Modules exchange typed signal objects containing sampled data, temporal information, and module-specific metadata.

Before execution, the canvas analyzer checks for cycles, unknown modules, missing mandatory inputs, isolated or unreachable nodes, incompatible signal types, and invalid parameter configurations. Cycle and reachability detection use standard graph traversal, while connection compatibility is determined from the typed module registry. A cycle is treated as invalid unless it passes through a module that explicitly introduces temporal state or delay.

After execution, numerical indicators and visual outputs are added to the canvas context. The result-analysis agent relates the observed outcomes to the experiment structure and parameter settings, explains abnormal phenomena, and recommends subsequent modifications or comparative trials. This closed inquiry process supports repeated construction, diagnosis, execution, observation, and refinement.

IV. EVALUATION AND DISCUSSION

A. Evaluation Setup

The framework was instantiated in an undergraduate Principles of Communication course covering modulation, channel capacity, noise, filtering, sampling, coding, bit-error-rate analysis, and spectrum interpretation. The tutoring evaluation comprised four controlled studies of KG enrichment, deterministic verification, multi-turn context handling, and learner-profile adaptation, totaling 64 system runs.

The experimentation system was evaluated on 20 construction, diagnosis, parameter-checking, and result-interpretation tasks under three configurations: a single-agent baseline (SA), a multi-agent configuration without canvas context (MA-NC), and the full multi-agent framework with canvas access (MA-FW). SA received only the learner request, MA-NC used specialist-agent orchestration without canvas information, and MA-FW additionally received the current modules, connections, parameters, diagnostics, and simulation results.

Within each comparison, the generation conditions used the

same underlying solver model and decoding settings. The generation order was randomized, and system identifiers were removed before evaluation. Responses were assessed using predefined rubrics and reference answers where applicable. Median latency is reported for the experimentation study because several runs contained high-latency outliers.

B. Tutoring Capability Evaluation

The same DeepSeek-based solver was used across the corresponding control and enabled conditions, while only the evaluated component or supplied context was changed. An independent Qwen3.7-Max model assessed answer quality, context resolution, and learner-profile compliance.

Table 1: Summary of intelligent tutoring capability results. Quality and compliance scores use a 0–100 scale; N/B/I/A denote neutral, beginner, intermediate, and advanced profiles.

Capability	Measurement	Value
KG enrichment	Mean quality (control/enabled)	98.5 / 93.5
Verification	Detection / final accuracy	100% / 90%
Multi-turn context	Resolution / accuracy (off/on)	50/100%; 50/83%
Learner profiles	Compliance (N/B/I/A)	78/34/84/58

KG retrieval was activated in all eight enabled runs and remained inactive in the control condition. However, mean judged quality decreased from 98.5 to 93.5, a mean paired difference of -5.0 points. The retrieval pipeline therefore affected generation but did not improve answers already handled well by the base model, indicating a need for more selective retrieval and context integration.

Deterministic verification detected all ten planted numerical errors and corrected nine, increasing final accuracy from 0% to 90% without introducing regressions. Multi-turn context increased reference resolution from 50% to 100% and answer accuracy from 50% to 83%. Learner profiles produced distinct responses for every question, but compliance remained uneven, particularly for beginner and advanced profiles. Verification and dialogue context therefore provided the clearest operational gains, whereas KG integration and profile-conditioned generation require further refinement.

C. Canvas-Grounded Experiment Evaluation

The experimentation system was assessed on four dimensions using a five-point rubric. Task completion measures whether the response addresses the requested experimental activity. Diagnostic quality measures the correctness and specificity of the identified problem and recommended action. Context grounding measures alignment with the supplied canvas state, while experimental pedagogy measures the quality of explanation, learner participation, and actionable next steps.

As shown in Fig. 2, task-completion scores remain similar across SA, MA-NC, and MA-FW, with values of 4.50, 4.50, and 4.55, respectively. The largest improvement occurs in context grounding, which increases from 3.85 for SA and 4.35 for MA-NC to 4.95 for MA-FW. Diagnostic quality also rises from 4.50

for SA and 4.60 for MA-NC to 4.85 for MA-FW. These results indicate that canvas access is most useful when guidance depends on the learner’s actual modules, connections, parameters, and signal-flow structure.

Experimental pedagogy increases from 4.15 for SA to 4.80 for both multi-agent configurations. Because MA-NC and MA-FW obtain the same score, this improvement appears to result mainly from specialist-agent orchestration, whereas canvas access contributes more strongly to grounding and diagnostic specificity. The additional capabilities also increase median latency from 8.10 s for SA to 17.45 s for MA-NC and 19.20 s for MA-FW, suggesting that most of the overhead originates from multi-agent coordination rather than canvas processing.

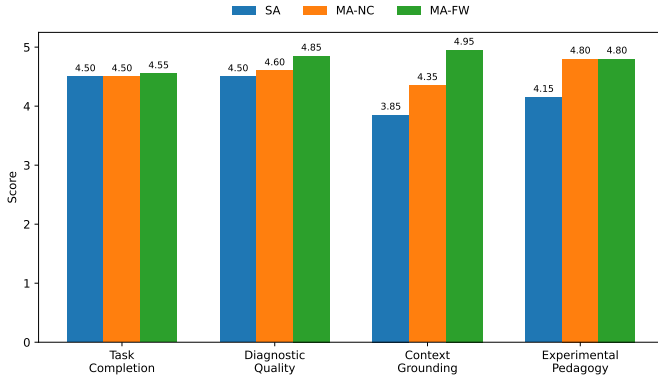


Figure 2: Comparison of the three system configurations on canvas-grounded experiment tasks. Scores are reported using a five-point rubric.

D. Discussion and Limitations

The results show that the framework components provide different forms of operational benefit. Deterministic verification improves numerical reliability, and dialogue context supports more accurate handling of follow-up questions. In the experimentation system, multi-agent orchestration mainly improves pedagogical organization, while canvas access strengthens contextual grounding and diagnostic specificity. In contrast, KG retrieval does not improve average quality on the current question set, and learner-profile compliance remains inconsistent. Complex workflows should therefore be activated selectively according to task requirements.

The study evaluates system capability and response quality rather than student learning outcomes and does not establish improvements in achievement, retention, or long-term learning. The framework was also evaluated only in a Principles of Communication implementation, so its transferability to other STEM domains remains to be validated. Furthermore, the canvas study evaluates generated guidance relative to the supplied experiment state rather than autonomous experiment repair. Future evaluation should include larger task sets, multiple human raters, classroom studies, canvas states before and after intervention, ground-truth fault labels, execution status, and simulation outputs.

V. CONCLUSION

This paper presented a domain-adaptable dual-system multi-agent framework for personalized tutoring and canvas-grounded experimentation. The tutoring evaluation showed that deterministic numerical verification and multi-turn context handling provided the clearest operational gains, whereas KG enrichment and learner-profile adaptation require more selective control. In the experimentation system, multi-agent orchestration mainly improved pedagogical organization, while canvas access strengthened contextual grounding and diagnostic specificity.

From an educational perspective, the framework helps connect conceptual explanation with experiment construction, observation, diagnosis, and reflection. Its value lies not only in generating answers, but also in supporting formative feedback that is responsive to the learner’s current question and experimental state. By combining specialist-agent guidance with learner checkpoints and actionable next steps, the framework is intended to preserve student participation rather than replace the inquiry process. The results also suggest that complex multi-agent workflows should be activated selectively, because their pedagogical and contextual benefits are accompanied by increased response latency.

The present study evaluates system capability in a Principles of Communication implementation rather than student learning outcomes or cross-domain generalization. Future work will examine classroom learning effects, teacher and student perceptions, execution-level experiment correction, and transfer to additional engineering and STEM domains.

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